

Specifications of MEMS sensors

Resolution. It is the smallest signal that a sensor can detect in the presence of noise. It is measured in 'g' (mg, ug).

Range. It refers to a maximum value of a signal that a sensor can detect. It is also measured in 'g'.

Bandwidth. It is the frequency range for which the output signal is within a specified deviation, typically $\pm 5\%$. It is measured in hertz.

Sensitivity. The change in output voltage of a sensor per unit change in input acceleration. It is measured in volts/g or mV/g.

Phase shift. It is the delay between the mechanical input and the corresponding electrical output signal.

Other parameters—temperature, humidity, electromagnetic noise tolerances, sensor mounting options, mounted resonant frequency, transverse or cross-axis sensitivity, mechanical resistance and dimensions.

MEMS in India

"Through a national program, Indian Institute of Science (IISc), Bangalore along with the Institute of Smart Structures and Systems (ISSS) has launched a major manpower development program in the emerging area of MEMS technology. National MEMS Design centres will be established in about 60 academic institutes across the country, and all the four major MEMS design tools, namely, Coventorware, Intellisuite, COMSOL and SoftMEMS will be installed at the design centres for student training and research. Further, MEMS technology course will be introduced in the curriculum for undergraduate and postgraduate students" as stated by Prof. Navakant Bhat, IISc Bangalore. "There are four foundries in India which can produce Silicon MEMS devices: SCL Chandigarh, BEL Bangalore, CEERI Pilani and Sitar Bangalore."

carefully analyse various parameters such as the number of axes, sensitivity, full scale, bandwidth, footprint and of course the price. Product designers have to trade off between bandwidth and resolution as stated by Prof. Anathasuresh, Indian Institute of Science, Bengaluru. Both gyroscopes and accelerometers are needed to accurately interpret the changes in linear and rotational motion as stated in its selection criterion by InvenSense.

"As more and more applications begin to use MEMS motion sensors, the requirements of these applications become equally diverse. For example, automobile airbag safety systems require high precision and very reliable high-g accelerometers, while cellphones require low-g accelerometers with less precision but very low power and a small package size," says Rob O'Reilly, director of business development, MEMS business, Analog Devices. The product designer needs to analyse the requirements and make a proper selection of the MEMS sensor.

Navigation. Motion sensing was

first put to use in the navigation system of an aircraft. The main elements of a navigation system involving motion are orientation, velocity and position. All these are efficiently provided by a combination of accelerometers and gyroscopes. Accelerometers detect the slope gradient by measuring the change in gravity, whereas gyroscopes give the accurate position. Navigation systems are used in aircrafts, in the in-built GPS systems of vehicles as well as external navigators. GPS is now a common component of smartphones and PDAs.

Gaming. Gaming involves a lot of control actions that are triggered by different motions. Many gaming consoles use motion sensors to control the gaming characters, creating a more interactive user interface. The future of the gaming industry is based on more interactive user interfaces and innovative use of various controls.

The most popular gaming controller, Nintendo's Wii, works on InvenSense's MEMS rate gyroscopes. The sensor measures the rate of rotation

along the three axes. The Wii's old controller could detect motions in different directions but could not distinguish the smaller or larger movements. The new MotionPlus version, based on rate gyroscopes, has this added functionality.

Mobile handsets. Apple's iPhone, Nokia, HTC, LG, Sony Ericsson and almost all smartphones have MEMS-based accelerometers as an integral part of their hardware design. Software applications that use the signals from the motion sensing microchip have converted the conventional mobile handset into a powerful 3D intelligent device. Motion sensing has become the 'must-have' function in mobile phones and PDAs.

Nokia's N95 was the first to have an accelerometer chip, but Apple's iPhone made the application popular. Another feature that's now common across mobiles, camcorders and cameras is the change of orientation of the display screen. Accelerometers and gyroscopes very accurately give the information on the orientation and inclination, which helps in accurately locating the position. This helps in reliable use of the GPS and navigation system. Gestures can be put to applications like gaming control, dialling numbers or changing the screen and scrolling. Recently, a news report described a Samsung phone that allows the user to dial the number by writing in the air. Mobile phones with cameras use accelerometers for image stabilisation and auto-focus features.

Medical Equipment. Pressure sensing is a big feature for numerous medical applications. BioMEMS is being talked about as another promising area in micro-fabrication and microfluidics technologies. MEMS sensors can sense the motion in context and take appropriate action. Their miniaturisation, accuracy and reliability are the key features that make MEMS suitable for medical implant devices like pacemakers. Motion sensors are used for invasive as well as non-invasive blood pressure measurement. Sensors can be embedded on a watch